

Thermal Modeling & Energy Optimization in Cloud Data Centers

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J. Lin, W. Lin, H. Huang, W. Lin and K. Li, "Thermal Modeling and Thermal-Aware Energy Saving Methods for Cloud Data Centers: A Review," in IEEE Transactions on Sustainable Computing, vol. 9, no. 3, pp. 571-590, May-June 2024

Why This Problem Matters?

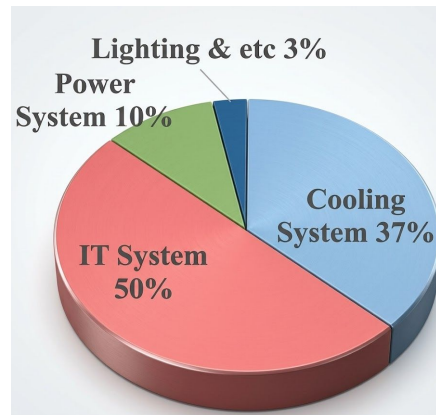
Data centers → rapidly growing energy demand

Overheating → performance loss + hardware risk

Cooling inefficiency → massive energy waste



Data center energy breakdown



The key issue is not just computing power, but the cost of keeping systems cool. Cooling alone can consume up to 40% of total energy.

Core Problem & Perspective

Workload



Heat generated by IT workload

Heat



Cooling must match dynamic thermal conditions

Model



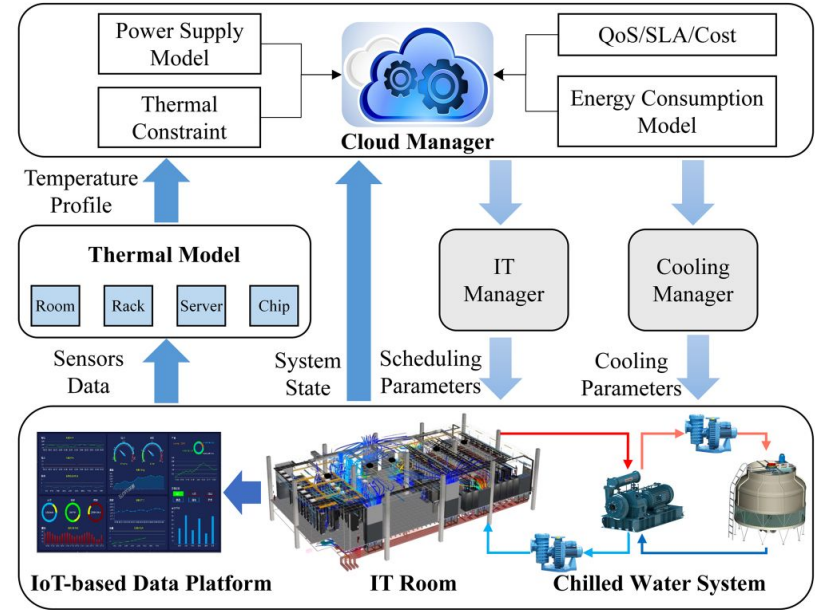
Thermal modeling and **Energy optimization**

Control

The challenge is balancing two things: avoiding overheating and minimizing energy consumption

Thermal Management Framework in Data Centers

- Sensors (IoT) collects **real-time data**
- Thermal model predicts **temperature**
- Controller optimizes:
 - **workload** distribution
 - **cooling** operation



This forms a feedback loop where decisions continuously adapt to system conditions.

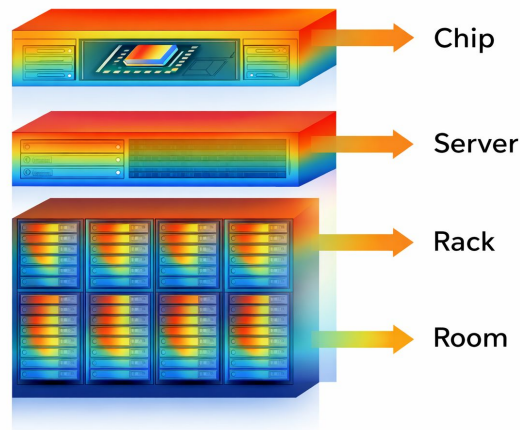
Cooling Architecture Evolves With Power Density

Low	Medium	High
Air cooling	Rack / row cooling	Liquid cooling
- raised floor, CRAC/CRAH, hot/cold aisle - limited by airflow mixing and recirculation	- shorter cooling path - better airflow predictability	- cold plate, immersion, spray - suited for high-density racks

The paper shows that cooling is **not one fixed architecture**. As rack **power density increases**, the system must move from coarse airflow-based cooling toward **more localized and efficient heat removal**.

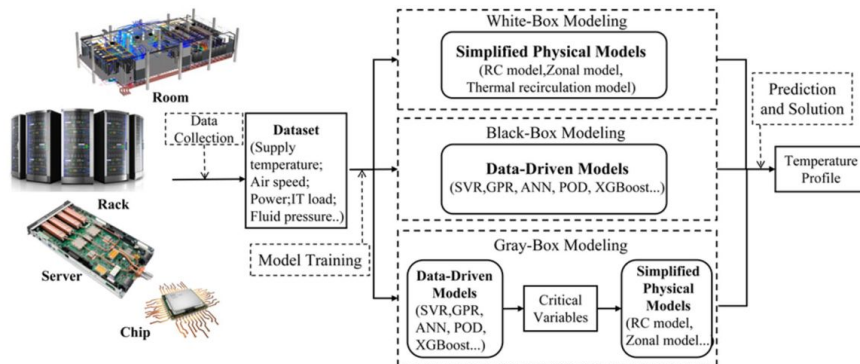
Thermal Modeling Is the Prediction Layer

- Thermal state is:
 - dynamic
 - nonlinear
 - spatially coupled
 - time-delayed
- Key prediction targets:
 - CPU / chip temperature
 - server inlet temperature
 - rack / room thermal profile
- Different decisions require different thermal granularity



Prediction accuracy determines control effectiveness

Three Modeling Paradigms



White-box

derived from thermodynamics
and heat transfer

Black-box

learned from sensor or
simulation data

Gray-box

uses physical structure with
data-estimated parameters

Model Trade-off

White-box	Black-box	Gray-box
physically interpretable	fast prediction	balances structure and adaptability
suitable for mechanism analysis	scalable with data	more practical for operational control
high modeling and computational cost	sensitive to training data quality	

The modeling problem is really a trade-off problem. For real deployment, the best model is not necessarily the most detailed one, but the one that is accurate enough, fast enough, and robust enough for control

Thermal-Aware Optimization Happens at Three Levels

IT load scheduling

place or migrate workloads
to reduce hotspots

Cooling control optimization

adjust setpoints, airflow,
or cooling capacity

Joint optimization

coordinate workload and
cooling decisions together

Thermal information becomes valuable only when translated into control decisions

Toward Intelligent Closed-Loop Thermal Control

Main challenges:

- thermal coupling across nodes
- sensor uncertainty and limited observability
- real-time control constraints
- energy vs performance trade-off

Future direction:

- AI-assisted prediction
- autonomous control
- tighter IT-cooling integration

Thermal management is a cyber-physical control problem